Guide to Session 1

Ubiquitous, Yet Hard to See · BCH 3023 · protein NMR: into a 1.2 GHz magnet, and reading what it gives

In Session 1 you put three samples of the protein ubiquitin into a 1.2 GHz NMR magnet, run the experiments NMR spectroscopists use on proteins, and read from each spectrum what it shows, and why it was so hard to get. This guide takes you through every step: **what to do**, **what you should see**, and **why** the step matters in a real NMR hall. Read the prelab review first (linked under the lab page's title).

Before you start

- **Your code** is at the top of the page. It gives you your own protein mass, residues, sequence, mystery sample, amides and stretch to work on. Use the same computer and browser throughout: the page keeps your work there.
- Every step can be done in the 3D hall (click a tube, the stair, the top of the magnet) or with the buttons beside it. This guide names the buttons.
- Each experiment shows how long it would take on a real magnet; the hall plays it in a few seconds.
- Allow about an hour and a half: about 15 minutes at the magnet, the rest reading the spectra and answering.
- The pictures here were taken on one code; your residues and peaks will differ.

In a real high-field NMR hall

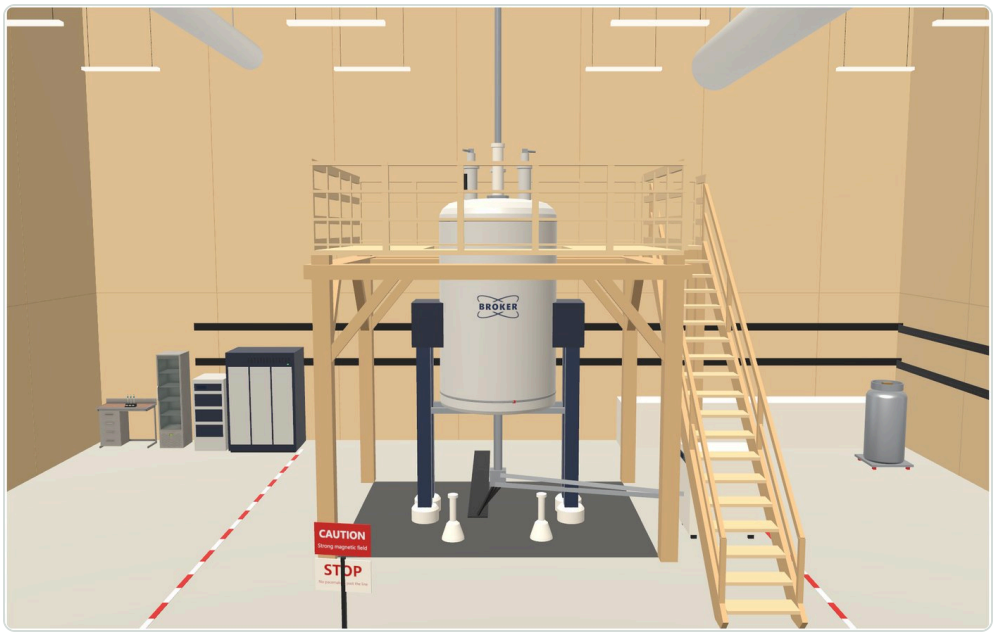
- **The 5 gauss line** on the floor (here, the red and white dashes) marks where the stray field may upset a pacemaker or another implanted device: people with one stay outside it unless the safety officer has cleared them.
- **Nothing of steel** goes inside the line: tools, gas cylinders, chairs and keys are pulled toward the magnet, harder the closer they get. Leave cards with magnetic stripes and watches outside.
- **Cryogens:** the magnet is kept cold with liquid helium and nitrogen. Only trained staff fill it, with a face shield and cryogenic gloves.
- **The oxygen monitor:** if the magnet quenches, its helium boils off. If the alarm sounds, leave at once.
- Climb the platform's stair one step at a time, with a hand on the rail and the tube in the other.

PART A

At the magnet

1 Look round the hall

The view opens on the whole hall: the magnet in the middle on its four legs, the timber platform round its top with the stair up the right side, the console cabinet (the spectrometer's electronics) and the cryoprobe's cooling unit beside the magnet, the sample bench at the back left and the desk at the front left. The buttons under the hall change the view: The hall | Sample bench | The magnet | Platform | Desk.

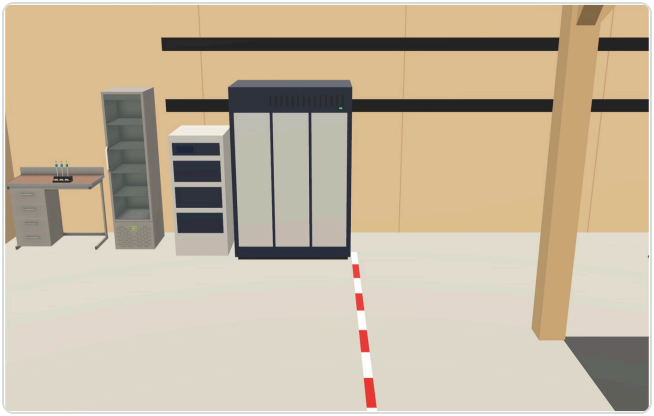


The NMR hall. The magnet's field is always on: the dashed line on the floor is the 5 gauss line.

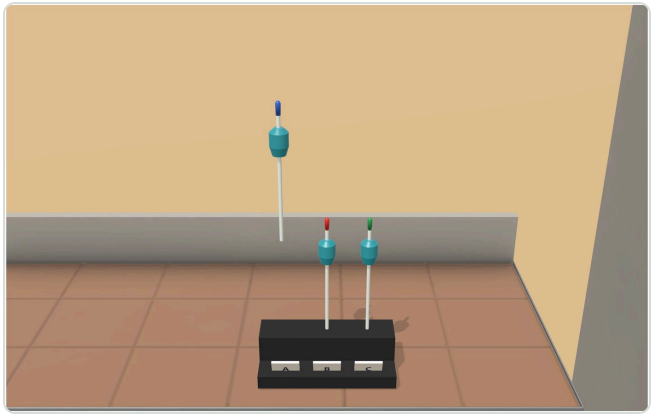
> **Why so big?** At 1.2 GHz, peaks a given distance apart in ppm are three times further apart in hertz than on a 400 MHz magnet, and the signal is stronger: a protein's crowded peaks need both.

2 Take sample A

Go to the sample bench: three NMR tubes stand in the rack, with their turquoise spinners on and their names, A, B and C, on the plates in front. Click tube A (or Take it beside Sample A in the list). You hold it above the rack.



The rack: A (blue cap), B (red), C (green).



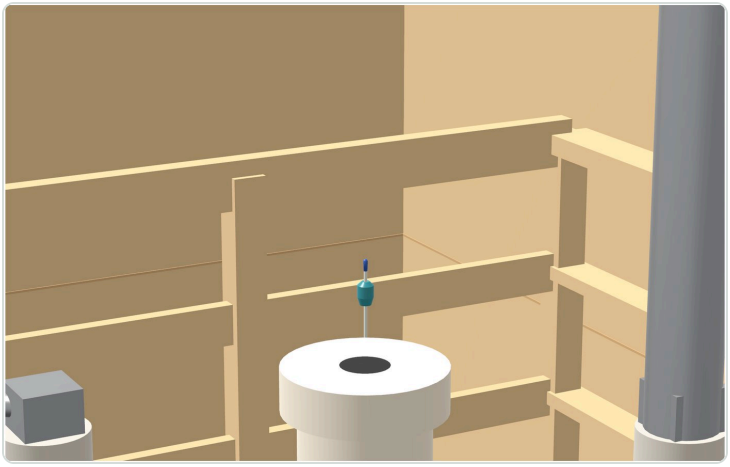
Tube A in hand.

The three samples. A: ubiquitin labelled with ^{15}N and ^{13}C, in 90% H_2O and 10% D_2O, for most experiments. B: the mystery sample, in 8 M urea and acid. C: ubiquitin just dissolved in D_2O, for the hydrogen exchange.

Why 10% D_2O? The spectrometer locks on the deuterium signal to hold its field steady; in 100% D_2O the amide hydrogens would trade for deuterium and their peaks would go.

3 Climb to the platform

Click the stair (or Climb the stair). You go up to the platform, beside the top of the magnet, with the tube held above the bore: the opening where the sample goes in.



On the platform: the tube waits above the bore, which is outlined in gold when the pointer is over it.

4 Lower the tube into the magnet

Click the top of the magnet (or Lower it in). The tube rides down the bore on a cushion of air, out of sight, to the probe at the magnet's centre, about two metres below.

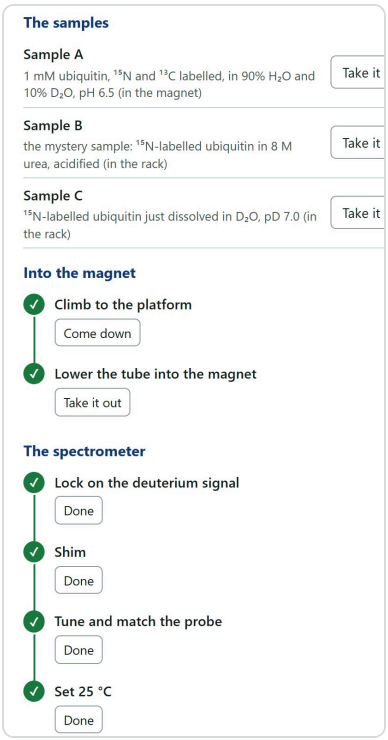


The tube is in: only the bore's opening shows.

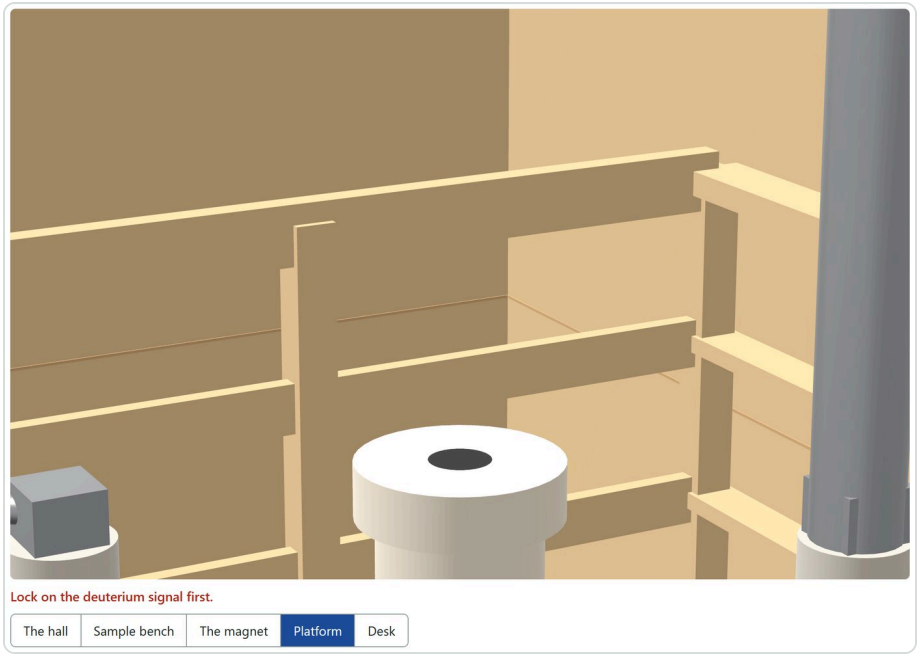
> Never let go of a tube over the bore without the air on: it would fall and break on the probe. (The hall does this for you.)

5 Set up the spectrometer

Under **The spectrometer**, in order: Lock, Shim, Tune, Set (25 °C). Each step's light turns into a green tick. The room refuses a step out of order and says why, in red under the hall.



The steps, all done for sample A.

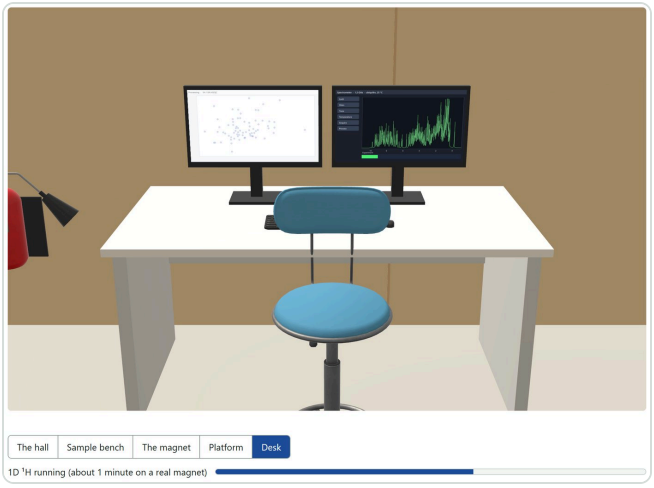


An experiment asked for before the lock: the room says what to do first.

> **Why each step:** the **lock** follows the solvent's deuterium to hold the field steady through runs of hours; **shimming** evens out the field over the sample so every line is sharp; **tuning and matching** fit the probe's circuits to this sample (salt changes them); and shifts move with temperature, so every spectrum is taken at 25 °C. All four must be done again for every new tube.

6 Run the experiments on sample A

Under **The experiments**, press Run beside each one meant for sample A, one at a time: the 1D ^{1}H, the 2D TOCSY, the 2D NOESY, the ^{1}H–^{15}N HSQC and the 3D HNCACB. While one runs, a bar fills under the hall and on the desk's screen, and the console cabinet's lamp blinks. When it is done, its spectrum opens in its part of the page, and a tick appears beside its name.



At the desk while the 1D runs.

The experiments

1D ^{1}H Sample A; about 1 minute	Run
2D TOCSY Sample A; about 2 hours	Run
2D NOESY Sample A; about 6 hours	Run
^{1}H–^{15}N HSQC Sample A or B; about 10 minutes	Run
3D HNCACB Sample A; about 4 hours with non-uniform sampling (about 2 days without)	Run
HSQCs in D_2O, over four weeks Sample C; 10 minutes each, the first as soon as the sample is in, the last after four weeks	Run

The times are typical for a 1 mM sample on a 1.2 GHz magnet with a cryogenically cooled probe; the room plays each in a few seconds.

Each experiment, its sample, and how long it takes on a real magnet.

> **Look at the times.** A 1D takes a minute; a 3D HNCACB takes about four hours even with non-uniform sampling (recording only part of the data and filling in the rest), about two days without. On a real protein project, weeks of magnet time go into the spectra you record here in a few minutes.

7 Change samples: B, then C

1. On the platform, Take it out (or click the top of the magnet): the tube rises back into your hand.
2. Come down (or click the stair), and Put it back in the rack.
3. Take sample B, climb, lower it in, and set up again: lock, shim, tune, 25 °C. Run the HSQC: it is your mystery spectrum.
4. Do the same with sample C, and run **HSQCs in D_2O, over four weeks**: the series of spectra for part 7.

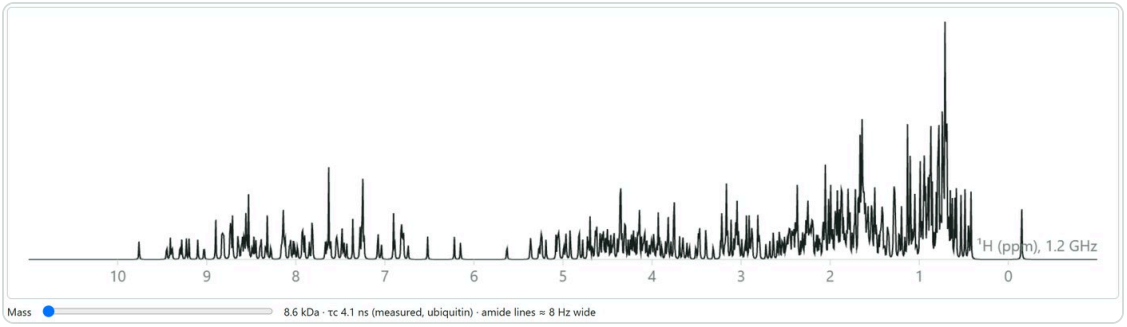
PART B

Reading your spectra

Each part of the lab page opens when its spectrum is recorded. The answers are checked as you type: a green "Accepted." or a hint in red. Your answers and your code print with the report.

8 The 1D proton spectrum

Ubiquitin's 600-odd protons, crowded between 0 and 10 ppm. Move the **Mass** slider: the page draws what a bigger protein would give, its lines broadened by slower tumbling, until the spectrum is one hump.

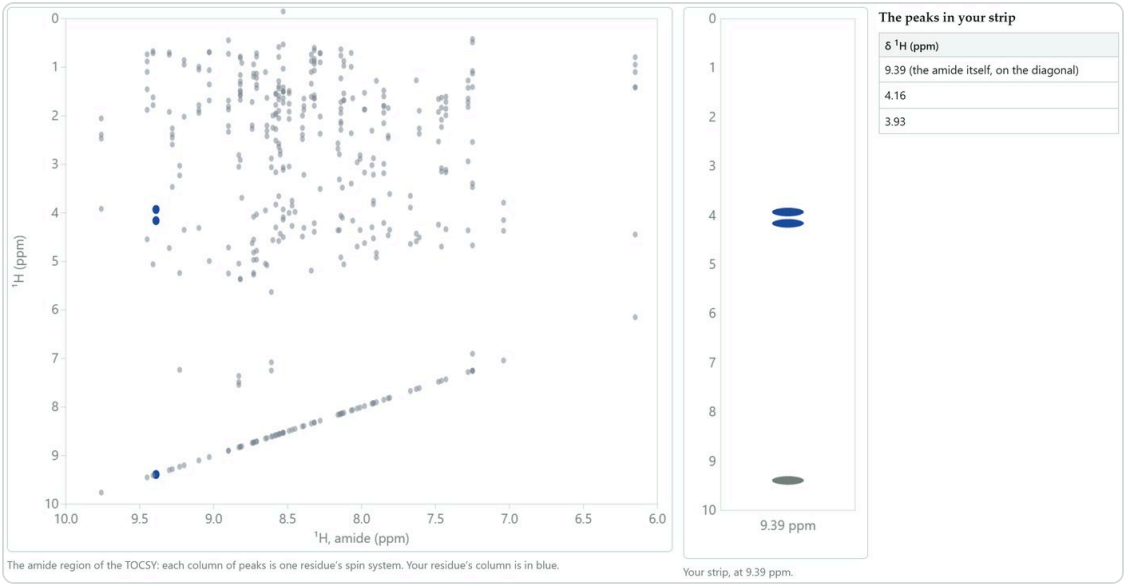


The 1D spectrum at ubiquitin's own mass.

To answer: use the rule of thumb on the page (about 0.6 ns of tumbling time per kDa) for your protein's mass. Then set the slider to your mass, read the line width it shows, and compare it with ubiquitin's.

9 TOCSY: name your residue

The whole amide region (left): each column of peaks is one residue's spin system. Your residue's column is in blue, and its strip and the table of its peaks are beside it.

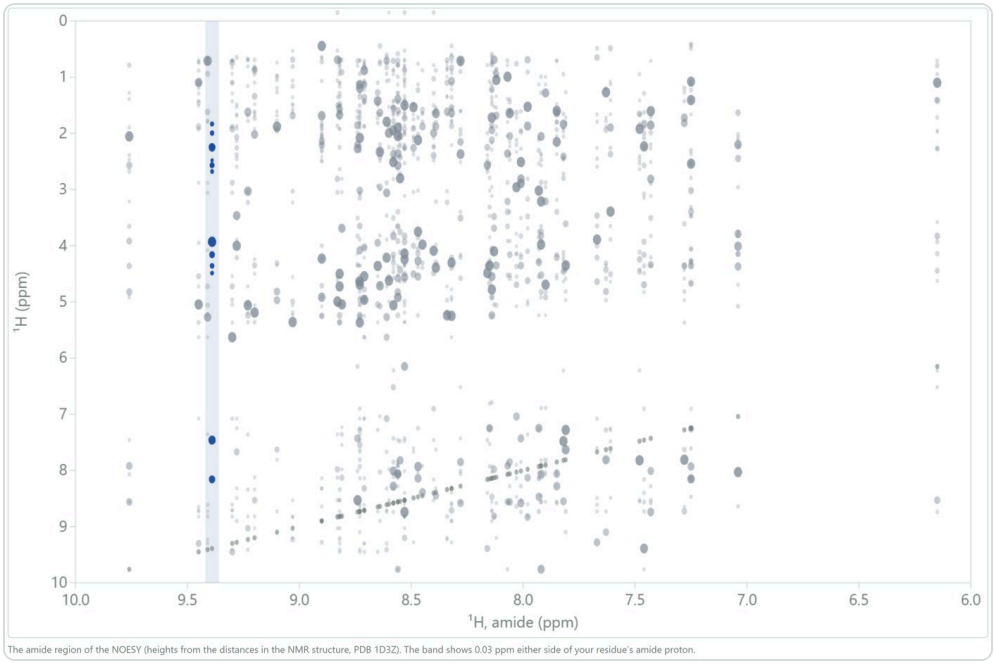


The TOCSY, the strip at one residue's amide proton, and its table of shifts.

To answer: open **Typical proton shifts of each amino acid** under the figure. Count your strip's peaks besides the amide; find the Hα (4–4.7 ppm); look for methyls ("a tall peak: three protons"), for protons between 3.5 and 4.5 ppm besides Hα (next to an OH), and for long chains. Match the pattern, not single numbers.

10 NOESY, and the crowd

Every proton within about 5 Å of each amide proton gives a peak. The blue band marks 0.03 ppm either side of your residue's amide proton.

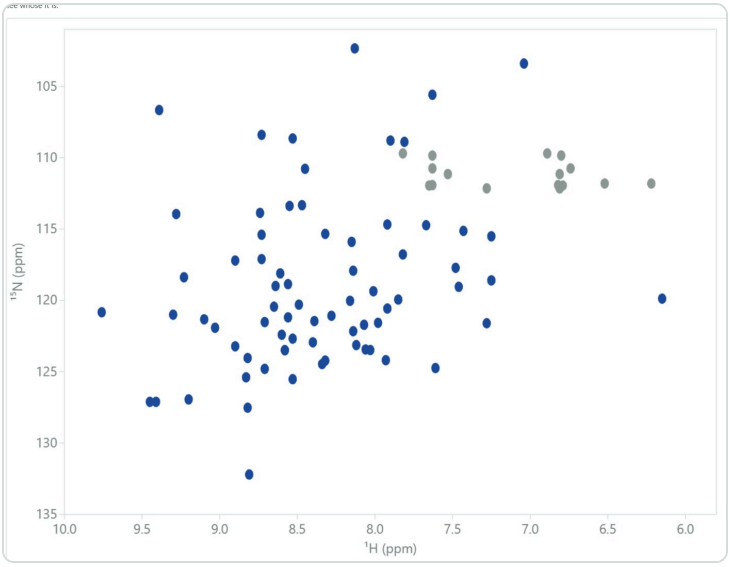


The NOESY's amide region, with the band at one residue.

To answer: count the other amide protons whose shift lies in the band. The HSQC's labels (point at its peaks) give every amide's exact shift. That count is how many columns a NOE at your shift could belong to.

11 The HSQC

One peak for each N–H: blue for the backbone amides, grey for the side-chain NH_2 of Asn and Gln (two peaks at one ^{15}N). Point at a peak to see whose it is and its shifts.

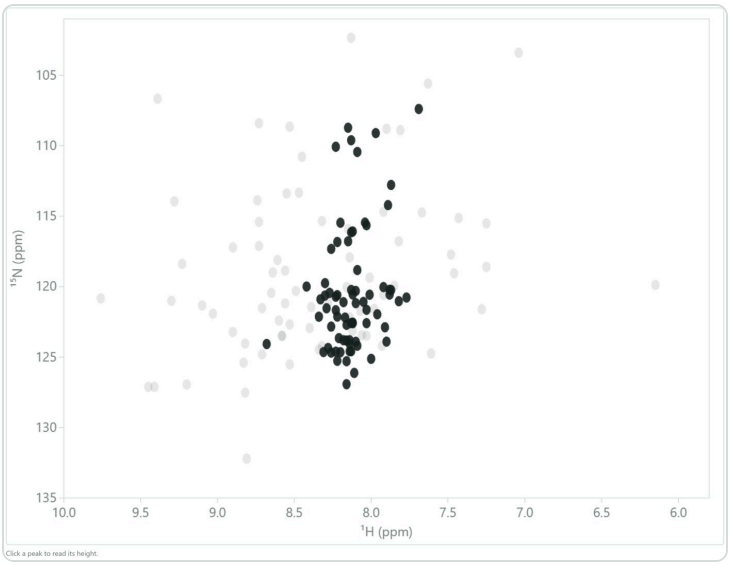


The ^{1}H–^{15}N HSQC.

To answer: for your made-up sequence, count the residues that give a backbone N–H (not the first, not prolines), then the side-chain peaks (two for each Asn and Gln, one for each Trp), then the total. The prelab works one such count.

12 Folded or unfolded?

Sample B's HSQC holds the folded protein's peaks and the unfolded protein's (crowded near 8.0–8.5 ppm) together, each set as tall as its share of the molecules.

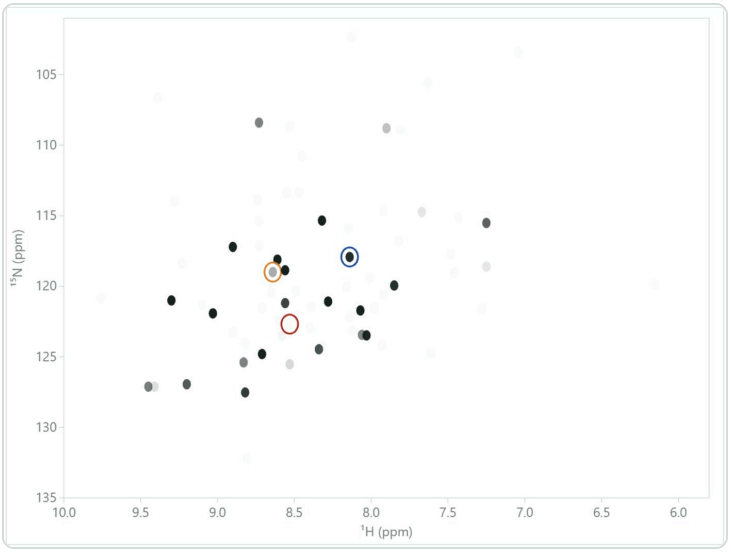


A mystery sample's HSQC (yours will differ).

To answer: click a peak of each kind to read its height (a peak of the fully folded protein is 1.00). The folded peaks' height is the folded share; the unfolded peaks', the unfolded share.

13 The D_2O shake

Choose a time under **Spectrum taken**: the HSQC taken then. Peaks fade as their amides trade H for D; your three amides are ringed (red, orange, blue).

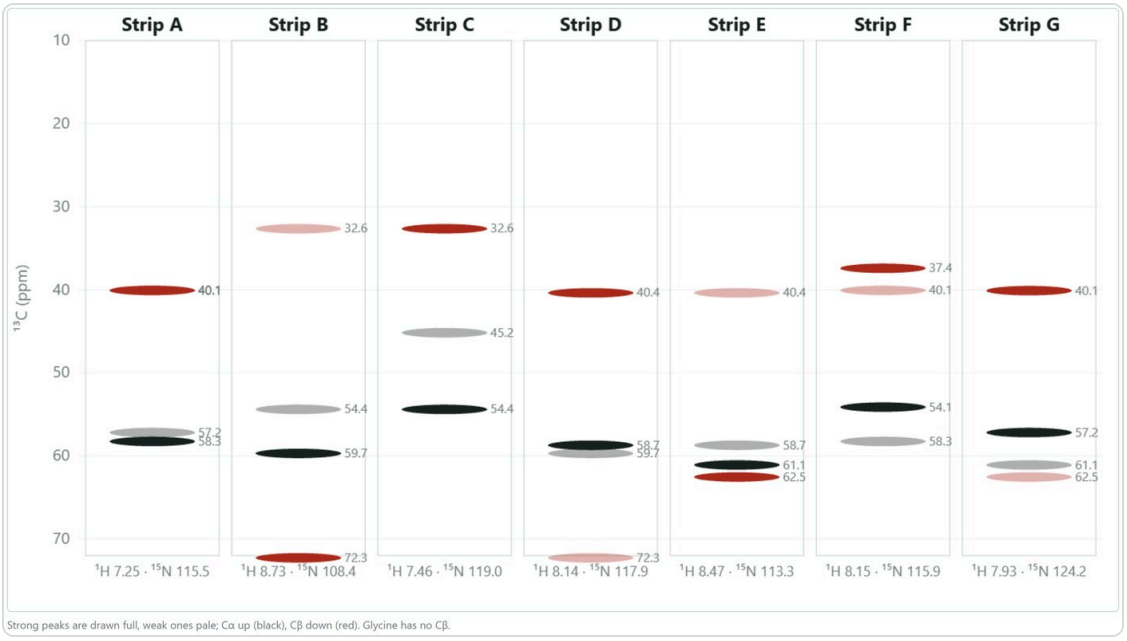


An HSQC six hours after the protein went into D_2O.

To answer: for each of your amides, click its peak in several spectra; each reading appears on the plot under the figure. Its half-life is the time at which its height is down to 0.50 (the plot has a line there). Then decide whether it is hydrogen-bonded: the slowest are the protected ones, and the structure under the table shows where each lies.

14 The sequential walk

Seven HNCACB strips, shuffled. Each shows its own residue's Cα (black, up) and Cβ (red, down) strongly, and the residue before's weakly.



Seven strips (yours will differ).

To answer: find the strip whose weak peaks match no strip's strong peaks: it comes first. Then, for each strip, find the one whose weak peaks match its strong peaks: that one comes next. Set the order in the boxes under the strips; when it is right, the strips are named.

15 Your report

Answer the three questions in your own words, type your name, and press Print the report, or save it as a PDF. The report prints every spectrum, every answer, your code, and what the page accepted. Upload the PDF where your instructor asks.

WHEN THE ROOM SAYS NO

The hall's messages

The message	What to do
An experiment is running: wait for it to finish.	Wait for the bar to fill.
A tube is still in the magnet: climb up and take it out first.	Climb the stair, Take it out, come down, put it back.
You already have a tube in hand: put it back first.	Put it back, then take the other.
The rack is on the sample bench: come down from the platform first.	Come down.
The top of the magnet is reached from the platform: climb the stair first.	Climb the stair.
There is no tube in the magnet.	Take a tube, climb, lower it in.

The message	What to do
Lock on the deuterium signal first (shimming follows the lock).	Lock, then the rest in order.
Shim the field first. / Tune and match the probe first. / Set the temperature to 25 °C first.	The step it names, under **The spectrometer**.
That experiment is not for the tube in the magnet.	See which sample each experiment is for: most are for A, the mystery HSQC for B, the D_2O series for C.

Quick reference

At the magnet

1. Sample bench: take tube A.
2. Climb the stair; lower it in.
3. Lock, shim, tune, 25 °C.
4. Run the 1D, TOCSY, NOESY, HSQC and HNCACB.
5. Take it out, come down, put it back.
6. Tube B: in, set up, run the HSQC (the mystery).
7. Tube C: in, set up, run the D_2O series.

Reading

8. 1D: τ_c from the rule of thumb; line widths from the slider.
9. TOCSY: your strip against the typical shifts.
10. NOESY: count amides within 0.03 ppm.
11. HSQC: count backbone, side-chain and total peaks.
12. Mystery: heights of folded and unfolded peaks.
13. D_2O: half-lives from the plot; hydrogen bonds.
14. Walk: chain the strips, weak to strong.
15. Report: three answers, your name, print to PDF.

The hall is a simulation: the magnet's maker, BROKER, and its name, Pinnacle, are made up. Every peak in the lab sits at a measured chemical shift of ubiquitin; line widths, heights and the fading in D_2O are modelled.